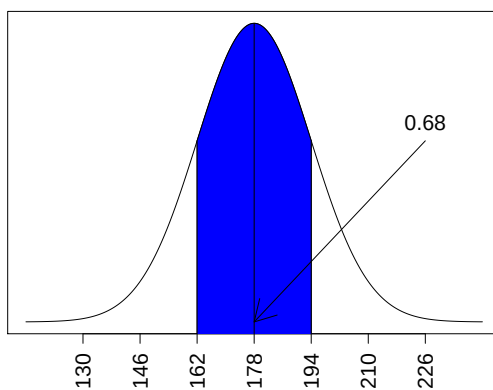


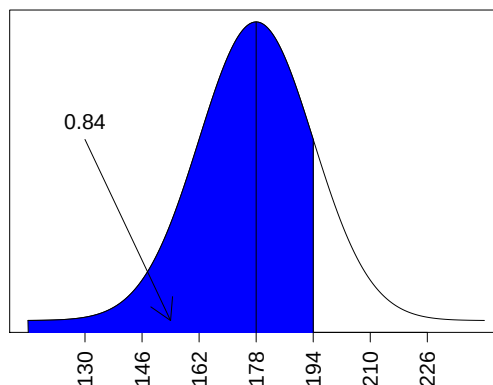
1. **Notation.** [7 pts] Suppose that a random variable, X , follows a Normal distribution. Specifically, $X \sim \mathcal{N}(-2, 16)$.
 - (a) What is the value of the mean? **-2**
 - (b) What is the value of the median? **-2**
 - (c) What is the value of the mode? **-2**
 - (d) What is the value of the standard deviation? **4**
 - (e) What is the value of the variance? **16**
 - (f) What is the total area under the density curve? **1**
 - (g) Multiple Choice. What is the range of values the random variable, X , can take on? (Enter the roman numeral corresponding to the correct answer, i.e. enter i, ii, iii, iv, or v.)
 - i. -14 to 10
 - ii. **$-\infty$ to ∞**
 - iii. -50 to 46
 - iv. -26 to 22
 - v. -98 to 94

2. **Corn.** [12 pts] Yield of corn (bushels per acre) in Nebraska counties in 2009 can be described by a Normal distribution with a mean of 178 bushels per acre and a standard deviation of 16 bushels per acre. Use **the 68-95-99.7 Rule (Empirical Rule)** to answer the following questions. (Hint: Draw pictures of the normal curve.)
 - (a) [4 pts] The middle 68% of counties have a corn yield **between** what two values?



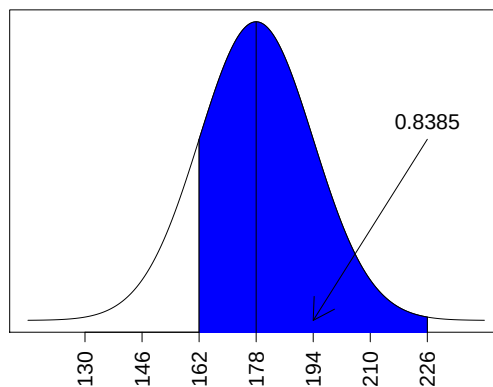
$$(\mu - \sigma, \mu + \sigma) = (178 - 16, 178 + 16) = 162 \text{ to } 194 \text{ bushels per acre}$$

- (b) [2 pts] What is the value of the 84th percentile of corn yield for counties in Iowa?



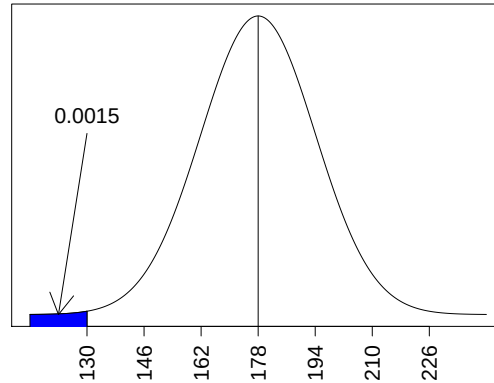
In the previous part, we found that the middle 68% of counties have a corn yield between 162 and 194 bushels per acre. This implies that $\frac{100-68}{2} = 16\%$ of counties have a corn yield greater than 194 bushels per acre which also implies 84% of counties have a corn yield less than 194 bushels per acre. So 194 bushels per acre is the value of the 84th percentile.

- (c) [2 pts] What proportion of counties have a corn yield **between** 162 and 226 bushels per acre?



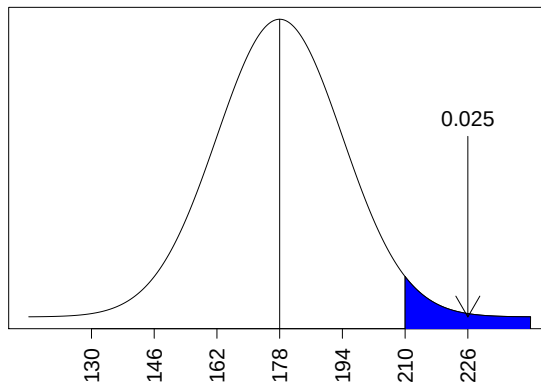
Previously, we found that the middle 68% of counties have a corn yield between 162 and 194 bushels per acre. This implies that $\frac{68}{2} = 34\%$ of counties have corn yields between 162 and 178 bushels per acre. 99.7% of counties have corn yields between $(\mu - 3\sigma, \mu + 3\sigma) = (178 - 48, 178 + 48) = (130, 226)$. This implies that $\frac{99.7}{2} = 49.85\%$ of counties have corn yields between 178 and 226 bushels per acre. Therefore, $34 + 49.85 = 83.85\%$ of counties have corn yields between 162 and 226 bushels per acre. So, the proportion of counties with corn yields between 162 and 226 bushels per acre is 0.8385.

- (d) [2 pts] 0.15% of counties have a corn yield **less than** or equal to what value?



99.7% of counties have corn yields between $(\mu - 3\sigma, \mu + 3\sigma) = (178 - 48, 178 + 48) = (130, 226)$. This implies the middle 99.7% of counties have corn yields between 130 and 226 bushels per acre. Therefore, 0.15% of counties have a corn yield less than 130 bushels per acre.

- (e) [2 pts] What proportion of counties have a corn yield of **at least** 210 bushels per acre?



95% of counties have corn yields between $(\mu - 2\sigma, \mu + 2\sigma) = (178 - 32, 178 + 32) = (146, 210)$. This implies the middle 95% of counties have corn yields between 146 and 210 bushels per acre. Therefore, 2.5% of counties have a corn yield greater than 210 bushels per acre. So, the proportion of counties with a corn yield of at least 210 bushels per acre is 0.0250.

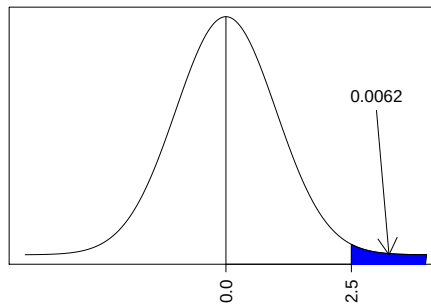
3. **Gas Mileage [27 pts]**. For a particular model of car, the miles per gallon (mpg) rating obtained for highway driving follows a Normal distribution with mean 28.5 mpg and standard deviation 1.4 mpg. Use this information to answer the following questions.

(a) **[3 pts]** What is the probability a car will obtain a miles per gallon rating for highway driving greater than 32 mpg? Answer this question by completing parts (i) and (ii).

i. **[1 pt]** Provide the z-score corresponding to the above statement.

$$z = \frac{x - \mu}{\sigma} = \frac{32 - 28.5}{1.4} = 2.50$$

ii. **[2 pts]** Based on your answer in (i), what is the probability a car will obtain a miles per gallon rating for highway driving greater than 32 mpg?



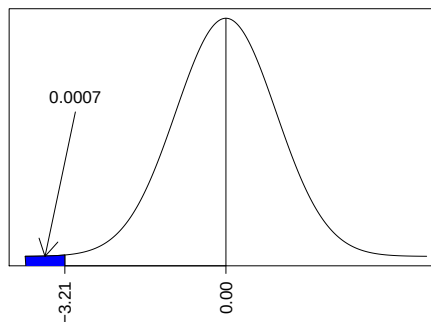
$$P(X > 32) = P(Z > 2.5) = 1 - P(Z < 2.5) = 1 - 0.9938 = 0.0062$$

(b) **[3 pts]** What is the probability a car will obtain a miles per gallon for highway driving less than 24 mpg? Answer this question by completing parts (i) and (ii).

i. **[1 pt]** Provide the z-score corresponding to the above statement.

$$z = \frac{x - \mu}{\sigma} = \frac{24 - 28.5}{1.4} = -3.21$$

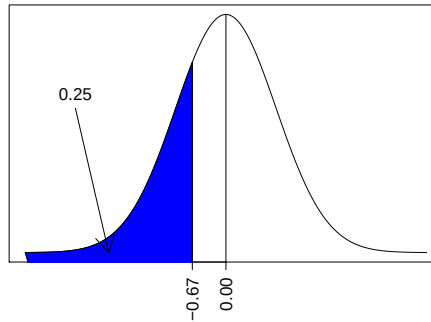
ii. **[2 pts]** Based on your answer in (i), what is the probability a car will obtain a miles per gallon for highway driving less than 24 mpg?



$$P(X < 24) = P(Z < -3.21) = 0.0007$$

(c) [3 pts] Find Q_1 (First Quartile). Answer this question by completing parts (i) and (ii).

i. [1 pt] Provide the z-score corresponding to the above statement.



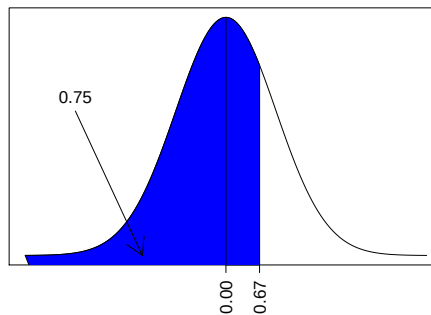
Q_1 is the 25th percentile. For a standard normal distribution, the 25th percentile is $z = -0.67$.

ii. [2 pts] Based on your answer in (i), provide the value of Q_1 . (Round answer to 3 decimal places.)

The 25th percentile of X is then $Q_1 = z \times \sigma + \mu = (-0.67)(1.4) + 28.5 = 27.562$ mpg.

(d) [3 pts] Find Q_3 (Third Quartile). Answer this question by completing parts (i) and (ii).

i. [1 pt] Provide the z-score corresponding to the above statement.



Q_3 is the 75th percentile. For a standard normal distribution, the 75th percentile is $z = 0.67$.

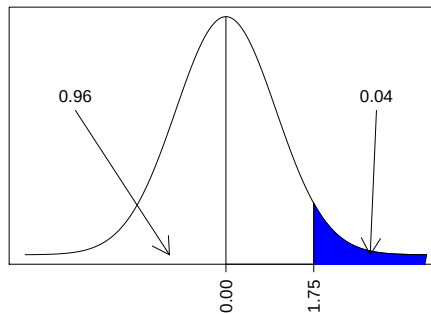
ii. [2 pts] Based on your answer in (i), provide the value of Q_3 . (Round answer to 3 decimal places.)

The 75th percentile of X is then $Q_3 = z \times \sigma + \mu = (0.67)(1.4) + 28.5 = 29.438$ mpg.

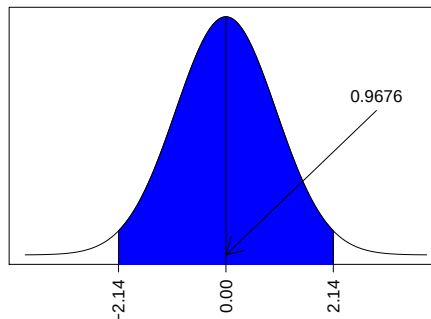
(e) [1 pt] What is the value of the IQR for the distribution of miles per gallon ratings for highway driving? (Round answer to 3 decimal places.)

$IQR = Q_3 - Q_1 = 29.438 - 27.562 = 1.876$ mpg.

- (f) [4 pts] What is the mpg rating such that only 4% of all cars have an mpg rating larger than that mpg value? Answer this question by completing parts (i) and (ii).
- i. [1 pt] Provide the z-score corresponding to the above statement.



- The z-score with 4% above it is the z-score with 96% below it. For a standard normal distribution, the z-score with 96% below it is $z = 1.75$.
- ii. [2 pts] Based on your answer in (i), what is the mpg rating such that only 4% of all cars have an mpg rating larger than that mpg value? (Round answer to 2 decimal places.)
 $x = z \times \sigma + \mu = (1.75)(1.4) + 28.5 = 30.95$ mpg.
- iii. [1 pt] What percentile does the mpg rating obtained in part (ii) correspond to? (Enter value as a whole number, not a proportion.)
 The 96th percentile.
- (g) [4 pts] What proportion of cars will have an mpg rating within 3 mpg of the mean? Answer this question by completing parts (i) through (iii).
- i. [1 pt] Provide the SMALLER z-score corresponding to the above statement.
 3 mpg below the mean is 25.5 mpg. So, $z = \frac{x - \mu}{\sigma} = \frac{25.5 - 28.5}{1.4} = -2.14$
- ii. [1 pt] Provide the LARGER z-score corresponding to the above statement.
 3 mpg above the mean is 31.5 mpg. So, $z = \frac{x - \mu}{\sigma} = \frac{31.5 - 28.5}{1.4} = 2.14$
- iii. [2 pts] Based on your answers in (i) and (ii), what proportion of cars will have an mpg rating within 3 mpg of the mean?



$$P(25.5 < X < 31.5) = P(-2.14 < Z < 2.14) = P(Z < 2.14) - P(Z < -2.14) = 0.9838 - 0.0162 = 0.9676$$

- (h) [3 pts] Which car is more unusual, a car with a 32 mpg rating for highway driving or a car with a 25 mpg rating for highway driving? Explain your answer.

Both cars are equally unusual because the mpg ratings corresponding to both cars are 3.5 mpg away from the mean which is 28.5 mpg.

- (i) [3 pts] Let's say you bought a car of the particular model discussed in this question and extensive testing shows that you get, on average, 28.5 mpg for your car on the highway. Provide an interpretation of your car's mpg rating, i.e. what can you say about how your car compares to all other cars of this particular model?

Since the mean is 28.5 mpg and the mpg rating of cars is Normal, we can say that we expect 50% of the cars to have a lower mpg rating as compared to our car while 50% of the cars will have a higher mpg rating compared to our car.

4. **SAT Scores.** [8 pts] A professor at a small university in New York surveys her introductory statistics students at the beginning of each semester. One of the questions she asks her students is what was the highest SAT score they received. The professor has collected data over many semesters and has SAT scores for 353 students. These data can be found on Blackboard (Assignments - Homework Assignments - Homework 2) in SAT.jmp/SAT.csv/SAT.sas/SAT.xlsx. Use your favorite software to create a histogram and a boxplot of the SAT scores as well as to calculate summary statistics. You do not need to give me the output, but you will need it to answer the following questions.

- (a) [3 pts] Based on the JMP output, describe the overall shape of the distribution of SAT scores for introductory statistics students that attended this small university in New York.

The distribution of SAT scores is unimodal, fairly symmetric, with no outliers.

- (b) [2 pts] Suppose you decide to use a Normal model to describe SAT scores for all introductory statistics students at this small university in New York. What possible values of the parameters (μ and σ) could you use for this Normal model?

The best guess we have for the values of the population parameters are the sample statistics. So, we would use the sample mean, $\bar{y} = 1206.72$ as an estimate for μ . (Note that since we are dealing with a symmetric, unimodal distribution with no outliers, you could also use the median=1200, but in general you want to estimate the population parameter with the same summary statistic from the sample.) Similarly, we would use the sample standard deviation, $s = 109.54$ as an estimate for σ .

- (c) [3 pts] The worst score a person can get on the SAT is a 400 and the best score a person can get is a 1600. A Normal model allows for any value, including values less than 400 and greater than 1600. Explain why it is still okay to use a Normal model to approximate the distribution of SAT scores.

The shape of the histogram (symmetric and unimodal with no outliers) provides evidence that a Normal model would be a good approximation to the distribution of SAT scores. While the Normal model allows for values from $-\infty$ to ∞ , we know that almost all values (99.7%) are within 3 standard deviations of the mean. Using our estimates for μ and σ from part (b), we know 99.7% of all observations would be between 878.10 and 1535.34. 400 is well below 878.10 and 1600 is greater than 1535.34, so for this particular Normal model, the probability of observing a value greater than 1600 or less than 400 is so small that the probability is essentially zero. This is why it is okay to use a Normal model to approximate the distribution of SAT scores.